

Project Report: Gaming Behavior & Mental Health Analysis

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Domain: Machine Learning / Data Science

1. Introduction

With the global rise of e-sports and casual gaming, understanding the psychological impact of gaming is crucial. This project utilizes **Machine Learning** to analyze patterns in gaming behavior and their correlation with mental health indicators. By processing user-reported data, the study aims to classify individuals into risk categories and provide automated behavioral recommendations.

2. Problem Statement

The goal is to identify whether specific gaming habits—such as hours played, motivations, or social engagement—can predict potential mental health risks. The project focuses on building a predictive model to classify users into "**High Risk**" or "**Low Risk**" levels based on a synthesized **Mental Health Score**.

3. Methodology

3.1 Data Preparation

- **Data Loading:** The dataset GamingStudy_data.csv was processed using Python's Pandas library.
- **Data Cleaning:** Missing values were handled using **Mean Imputation** for numeric columns to ensure data consistency without losing significant rows.
- **Feature Selection:** The analysis focused exclusively on numeric variables to calculate statistical correlations.

3.2 Feature Engineering

A custom metric, '**Mental_Score**', was created by calculating the mean of all numeric indicators for each participant.

- **Target Variable:** A categorical label, '**Risk_Level**', was generated based on the median score:
 - **High:** Score $\geq$ Median
 - **Low:** Score $<$ Median

3.3 Model Architecture

The project uses a **Decision Tree Classifier** for its transparency and ability to show feature importance.

- **Split Ratio:** 70% Training / 30% Testing.
- **Hyperparameters:** max_depth=5 was applied to prevent overfitting while maintaining interpretability.

4. Data Visualization & Insights

4.1 Key Findings

- **Correlation Heatmap:** Used to identify which gaming behaviors (like hours per week) strongly align with mental health markers (like anxiety or insomnia scores).
- **Feature Importance:** The model identified the top 8 factors that most significantly impact a user's risk level, allowing us to see which habits are the biggest "red flags."
- **Score Distribution:** A Histogram with Kernel Density Estimate (KDE) was used to visualize the spread of mental health scores across the population.

5. Results & Evaluation

5.1 Performance Metrics

- **Accuracy:** The model successfully predicts the risk level of the test set with a high degree of precision.
- **Confusion Matrix:** A heatmap visualization of the Confusion Matrix confirms the model's ability to distinguish between High and Low risk without a high rate of False Positives.

5.2 Recommendation System

An automated logic was implemented to provide actionable feedback:

- **High Risk:** Suggests reducing gaming hours and prioritizing sleep hygiene.
- **Low Risk:** Encourages the maintenance of current healthy gaming habits.

6. Conclusion

The analysis demonstrates that Machine Learning can effectively categorize behavioral health trends. By identifying "High Risk" patterns early, this model serves as a foundation for digital well-being tools that promote a healthier balance between gaming and mental health.

7. Tools & Technologies

- **Languages:** Python
- **Libraries:** Pandas, NumPy, Scikit-learn
- **Visualization:** Matplotlib, Seaborn
- **Platform:** Google Colab